

**UNIVERSITY OF PORT HARCOURT
POST-UTME/BASIC ENTRANCE EXAM**

1. 25cm^3 of a mixture of noble gases was obtained from air after the removal of other constituents. What was the original volume of air?
A. 125cm^3
B. 1250cm^3
C. 2500cm^3
D. 80cm^3
2. Which of the following gases will diffuse fastest when passed through a porous plug?
A. Propane
B. Oxygen
C. Methane
D. Ammonia
3. In which of the following are water molecules in the most disorderly arrangement?
A. Ice at -10°C
B. Ice at 0°C
C. Water at 100°C
D. Steam at 100°C
4. Nitrogen obtained from the liquefaction of air has a higher density than that obtained from nitrogen containing compounds because the former contains .
A. water vapour
B. oxygen
C. carbon (IV) oxide
D. rare gases.
5. The method that can be used to convert hard water to soft water is
A. chlorination
B. passage over activated charcoal
C. the use of an ion-exchange resin
D. aeration.
6. $1s^2 2s^2 2p^2 3s^2 3p^6 3d^7 4s^2$. An element with the electron configuration above is a
A. non-metal
B. metal
C. transition element
D. group two element.
7. The pair of metals in the reactivity series that are usually extracted by the electrolysis of their ores is
A. magnesium and zinc
B. magnesium and calcium
C. copper and zinc
D. lead and calcium
8. A metal that can be extracted from cassiterite is
A. calcium
B. magnesium
C. tin
D. copper
9. Which of the following is a solvent for perfumes?
A. C_5H_{12}
B. C_4H_6
C. CH_3COOH
D. 2-methylbut-1-ene.
10. The partial pressure of oxygen in a sample of air is 452mmHg and the total pressure is 780mmHg . What is the mole fraction of oxygen?
A. 5,790
B. 2.030
C. 0.579
D. 0.203
11. The knowledge of half-life can be used to
A. create an element
B. split an element
C. detect an element
D. irradiate an element
12. The noble gas, argon, is used for
A. steel welding
B. welding brass
C. underwater welding
D. electric arc welding
13. The electron configuration of ${}_{22}\text{X}^{2+}$ ion is
A. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$
B. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$
C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
D. $1s^2 2s^2 sp^6 3s^2 sp^6$
14. Which of the following types of bonding does not involve the formation of new substances?
A. Covalent
B. Metallic
C. Co-ordinate
D. Electrovalent
15. The shapes of CO_2 , H_2O and CH_4 respectively are
A. bent, linear and tetrahedral

- B. bent, tetrahedral and linear
C. tetrahedral, linear and bent
D. linear, bent and tetrahedral.
16. $\text{HCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})} \rightleftharpoons \text{H}_3\text{O}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})}$ in the reaction above, $\text{Cl}_{(\text{aq})}^-$ is the
A. conjugate base
B. acid
C. conjugate acid
D. base
17. $\text{MnO}_4^-_{(\text{aq})} + 8\text{H}^+_{(\text{aq})} + \text{Y} \rightarrow \text{Mn}^{2+}_{(\text{aq})} + 4\text{H}_2\text{O}_{(\text{l})}$
Y in the equation above represents
A. 7e^-
B. 5e^-
C. 3e^-
D. 2e^-
18. Powdered marble reacts faster with hydrochloric acid solution, than the granular form because the powdered form has
A. more molecules
B. larger surface area
C. more atoms
D. relatively large mass
19. The refreshing and characteristic taste of soda water and other soft drink is as a result of the presence in them of
A. soda
B. carbon (II) oxide
C. carbon (IV) oxide
D. glucose
20. Alkanones are generally obtained by the oxidation of
A. primary alkanols
B. tertiary alkanols
C. secondary alkanols
D. alkanic acid.
21. Sucrose is made up of
A. glucose and fructose
B. glucose and glucose
C. fructose and fructose
D. galactose and glucose
22. The chlorinated alkane often used industrially to remove grease is
A. chloromethane
B. tetrachloromethane
C. trichloromethane
D. dichloromethane
23. What mass of water is produced when 8.0 g of hydrogen reacts with excess oxygen?
A. 72.0g
B. 36.0g
C. 16.0g
D. 8.0g
24. Four elements W, X, Y, and Z have atomic numbers 2, 6, 16 and 20 respectively. Which of these elements is a metal?
A. X.
B. Z.
C. W.
D. Y.
25. An element X with relative atomic mass 16.2 contains two isotopes $^{16}_8\text{X}$ with relative abundance of 90% and ^m_8X with relative abundance of 10%. The value of m is
A. 14
B. 12
C. 18
D. 16
26. Cancerous growths are cured by exposure to
A. X-rays
B. β -rays
C. α -rays
D. γ -rays
27. Which of the following statements is correct about the average kinetic energy of the molecules of a gas?
A. It increases with increase in pressure
B. It increases with increase in temperature
C. It increases with increase in volume
D. It increases at constant pressure
28. The chemical used for coagulation in water purification is
A. copper tetraoxosulphate (VI)
B. sodium tetraoxosulphate (VI)
C. aluminium tetraoxosulphate (VI)
D. calcium tetraoxosulphate (VI)
29. What current in amperes will deposit 2.7g of aluminium in 2 hours?
A. 32
B. 16
C. 8

- D. 4
30. As the concentration of an electrolyte reduces, the conductivity
- decreases
 - increases
 - reduces to zero
 - is unaffected
31. The general formula for the alkanols is
- RCOOR
 - R₂CO
 - RCHO
 - ROH
32. The gas that can best be collected by downward displacement of air is
- chlorine
 - sulphur (IV) oxide
 - carbon (IV) oxide
 - ammonia
33. Which of the following equations shows that a reaction is in equilibrium?
- $\Delta G = 0$
 - $\Delta G > 0$
 - $\Delta G < 0$
 - $\Delta G = \Delta H - T\Delta S$
34. $\text{Cu}_2\text{S}_{(s)} + \text{O}_{2(g)} \rightarrow 2\text{Cu}_{(s)} + \text{SO}_{2(g)}$
What is the change in the oxidation number of copper in the reaction above?
- 0 to +1
 - 0 to +2
 - +2 to +1
 - +1 to 0
35. A compound contains 31.91% potassium, 28.93% chlorine and the rest oxygen. What is the chemical formula of the compound?
- KClO₃
 - KClO₄
 - KClO
 - KClO₂
36. The type of reaction that is peculiar to benzene is
- hydrolysis
 - addition
 - substitution
 - polymerization
37. How many isomers does pentane have?
- 4
- B. 3
- C. 5
- D. 6
38. The formula for ethyl butanoate is
- C₄H₉COOC₂H₅
 - C₂H₅COOC₄H₉
 - C₂H₅COOC₃H₇
 - C₃H₇COOC₂H₅
39. If a solution contains 4.9g of tetraoxosulphate (VI) acid, calculate the amount of copper (II) oxide that will react with it.
- 0.8g
 - 4.0g
 - 40.0g
 - 80.0g
40. Chlorine consisting of two isotopes of mass number 35 and 37 in the ratio 3:1 has an atomic mass of 35.5. Calculate the relative abundance of the isotope of mass number 37.
- 20
 - 25
 - 60
 - 75
41. Hardness of water is caused by the presence of the ions of
- magnesium and silver
 - calcium and magnesium
 - sodium and potassium
 - calcium and sodium
42. Alloys are best prepared by
- electroplating
 - arc-welding
 - reducing a mixture of their metallic Oxides
 - cooling a molten mixture of the metals
43. In the production of soap, concentrated sodium chloride solution is added to
- increase the solubility of the soap
 - decrease the solubility of the soap
 - saponify the soap
 - emulsify the soap
44. Which of these reagents can confirm the presence of a triple bond?
- Bromine water
 - Acidified KMnO₄

- C. Copper (I) Chloride
D. Bromine gas.
45. Sugar is separated from its impurities by
A. distillation
B. evaporation
C. precipitation
D. crystallization
46. The shape of the hydrocarbon compound CH_4 is
A. square planar
B. tetrahedral
C. planar
D. linear
47. If 30cm^3 of a gas at 50°C is warmed to 80°C at a fixed pressure the fractional increase in volume is
A. 0.093
B. 1.090
C. 0.009
D. 0.910
48. The soft solder used for welding and plumbing is an alloy of
A. lead and tin
B. iron and tin
C. lead and zinc
D. iron and zinc
49. How long would it take to deposit 0.08g of copper from CuCl_2 solution by passing a current of 0.5A?
A. 6 mins.
B. 8 mins.
C. 24 mins.
D. 48 mins.
50. The oxidation state of oxygen in tetraoxosulphate (IV) acid is
A. -4
B. -2
C. +4
D. -8
51. Which of the following represents an aromatic compound?
A. $\text{C}_6\text{H}_5\text{OH}$
B. C_6H_{12}
C. $\text{C}_6\text{H}_9\text{OH}$
D. C_6H_8
52. Which of the following metals is purified commercially by electrolysis?
A. Zn
B. Fe
C. Sn
D. Cu
53. The presence of nitrogen in air is to slow down
A. respiration and combustion
B. combustion and corrosion
C. corrosion and respiration
D. respiration and transpiration
54. Bronze is preferred to copper in the making of medals because it
A. is stronger
B. can withstand low temperature
C. is lighter
D. has a low tensile strength
55. How many faraday of electricity is required to produce 0.25mol of copper?
A. 0.01F
B. 0.05F
C. 0.50F
D. 1.00F
- Mathematics**
1. Evaluate $\log_{\sqrt{2}}4 + \log_{\frac{1}{2}}16 - \log_432$
A. -5.5
B. -2.5
C. 2.5
D. 5.5
2. In a class of 40 students, 32 offer Mathematics, 24 offer Physics and 4 offer neither Mathematics nor Physics. How many offer both Mathematics and Physics?
A. 20
B. 16
C. 8
D. 4
3. Find $\left(\frac{1}{0.06} + \frac{1}{0.042}\right)^{-1}$, correct to two decimal places
A. 1.43
B. 1.53
C. 3.14
D. 4.42
4. A woman buys 270 oranges for ₦1800.00 and sells at 5 for ₦40.00. What is her profit?
A. ₦2 160.00
B. ₦1 620.00
C. ₦630.00

- D. ~~N~~360.00
5. Simplify $1 - \left(2\frac{1}{3} \times 1\frac{1}{4}\right) + \frac{3}{5}$
- A. $-1\frac{1}{15}$
 B. $-1\frac{19}{60}$
 C. $-2\frac{7}{15}$
 D. $-2\frac{31}{60}$
6. A cinema hall contains a certain number of people. If $12\frac{1}{2}\%$ are children, $47\frac{1}{2}\%$ are men and 84 are women, find the number of men in the hall.
- A. 63
 B. 84
 C. 113
 D. 133
7. If $\frac{9^{2x-1}}{27^{x+1}} = 1$, find the value of x.
- A. 8
 B. 5
 C. 3
 D. 2
8. Factorize completely $4abx - 2axy - 12b^2x + 6bxy$
- A. $2x(a-3b)(2b-y)$
 B. $2x(3b-a)(2b-y)$
 C. $2x(a-3b)(y-2b)$
 D. $2x(2b-a)(3b-y)$
9. An arc of a circle subtends an angle of 30° on the circumference of a circle of radius 21cm. Find the length of the arc.
- A. 11cm
 B. 22cm
 C. 44cm
 D. 66cm
10. A trapezium has two parallel sides of length 5cm and 9cm. if the area is 21cm^2 , find the distance between the parallel sides.
- A. 3cm
 B. 4cm
 C. 6cm
 D. 7cm
11. BY how much is the mean of 30, 56, 31, 55, 43, and 44 less than the median?
- A. 0.75
 B. 0.50
 C. 0.33
 D. 0.17
12. On a pie chart, there are four sectors of which three angles are 45° , 90° and 135° . If the smallest sector represents ~~N~~28.00, how much is the largest sector?
- A. ~~N~~96.00
 B. ~~N~~84.00
 C. ~~N~~48.00
 D. ~~N~~42.00
13. Two dice are thrown. What is the probability that the sum of the numbers is divisible by 3?
- A. $\frac{2}{3}$
 B. $\frac{1}{2}$
 C. $\frac{1}{3}$
 D. $\frac{1}{4}$
14. The mean age of a group of students is 15 years. When the age of a teacher, 45 years old, is added to the ages of the students, the mean of their ages becomes 18 years. Find the number of students in the group.
- A. 9
 B. 7
 C. 42
 D. 15
15. Given that the first and fourth terms of a G.P. are 6 and 162 respectively, find the sum of the first three terms of the progression.
- A. 27
 B. 8
 C. 78
 D. 48
16. If the operation on the set of integers is defined by $p.q = \sqrt{pq}$, find the value of 4. (8.32).
- A. 8
 B. 3
 C. 16
 D. 4
17. Factorize completely $ac-2bc - a^2 + 4b^2$.
- A. $(a-2b)(c-a-2b)$
 B. $(a-2b)(c+a+2b)$
 C. $(a-2b)(c-a+2b)$
 D. $(a-2b)(c+a-2b)$

18. Find the sum to infinity of the series $\frac{1}{2}, \frac{1}{6}, \frac{1}{18}$
- A. $\frac{2}{3}$
 B. $\frac{1}{3}$
 C. $\frac{3}{4}$
 D. 1
19. If $6\log_x 2 - 3\log_x 3 = 3\log_{50} 2$, find x.
- A. $\frac{8}{3}$
 B. $\frac{4}{3}$
 C. $\frac{3}{4}$
 D. $\frac{3}{8}$
20. Evaluate $\frac{\frac{1}{10} \times \frac{2}{3} + \frac{1}{4}}{\frac{1}{2} + \frac{3}{5} \times \frac{1}{4}}$
- A. $\frac{7}{12}$
 B. $\frac{19}{35}$
 C. $\frac{2}{25}$
 D. $\frac{19}{60}$
21. Simply $\frac{1}{\sqrt{3+2}}$ in the form a + b
- A. $2 - \sqrt{3}$
 B. $-2 - \sqrt{3}$
 C. $2 + \sqrt{3}$
 D. $-2 + \sqrt{3}$
22. A number is selected at random between 20 and 30 both numbers inclusive. Find the probability that the number is a prime
- A. $\frac{2}{11}$
 B. $\frac{5}{11}$
 C. $\frac{6}{11}$
 D. $\frac{8}{11}$
23. What is the nth term of the sequence 2, 6, 12, 20, ...?
- A. $4n - 2$
 B. $2(3n - 1)$
 C. $n^2 + n$

- D. $n^2 + 3n + 2$
24. Simply $3\frac{1}{2} - \left(2\frac{1}{3} \times 1\frac{1}{4}\right) + \frac{3}{5}$
- A. $2\frac{11}{60}$
 B. $2\frac{1}{60}$
 C. $1\frac{11}{60}$
 D. $1\frac{1}{60}$
25. Find the value of m if $13_m + 24_m = 41_m$
- A. 8
 B. 6
 C. 5
 D. 2
26. If the interest on ₦150.00 for $2\frac{1}{2}$ years is 4.50, find the interest on N250.00 for 6 months at the same rate.
- A. ~~₦~~1.50
 B. ~~₦~~7.50
 C. ~~₦~~15.00
 D. ~~₦~~18.00
27. Three boys shared some oranges. The first received $\frac{1}{3}$ of the oranges and the second received $\frac{2}{3}$ of the remainder. If the third boy received the remaining 12 oranges, how many oranges did they shared?
- A. 60
 B. 54
 C. 48
 D. 42
28. Evaluate $\frac{81^{\frac{3}{4}} - 27^{\frac{1}{3}}}{3 \times 2^3}$
- A. 3
 B. 1
 C. $\frac{1}{3}$
 D. $\frac{1}{8}$
29. If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, evaluate $\log_{10} 4.5$
- A. 0.9542
 B. 0.6532
 C. 0.4771
 D. 0.3010

30. Simplify $\frac{\sqrt{12}-\sqrt{3}}{\sqrt{12}+\sqrt{3}}$
 A. 0
 B. $\frac{1}{3}$
 C. $\frac{3}{5}$
 D. 1
31. If $m = 3$, $p = -3$, $q = 7$ and $r = \frac{5}{2}$, evaluate $m(p + q + r)$.
 A. 19.50
 B. 19.15
 C. 18.95
 D. 18.05
32. The weight Wkg of a metal bar varies jointly as its length L metres and the square of its diameter d metres. If $W = 140$ when $d = 4\frac{2}{3}$ and $L = 54$, find d in terms of W and L
 A. $\sqrt{\frac{42W}{5L}}$
 B. $\sqrt{\frac{5L}{42W}}$
 C. $\sqrt{\frac{42W}{5L}}$
 D. $\sqrt{\frac{5L}{42W}}$
33. If the 7th term of an AP is twice the third term and the sum of the first four terms is 42, find the common difference.
 A. 6
 B. 3
 C. 2
 D. 1
34. Find the sum of the first 20 terms of the series 8, 12, 16..., 96.
 A. 1400
 B. 1040
 C. 960
 D. 920
35. If the mean of five consecutive integers is 30, find the largest of the numbers.
 A. 28
 B. 30
 C. 32
 D. 34
36. A bag contains 5 black, 4 white and x red marbles. If the probability of picking a red marble is $\frac{2}{5}$, find the value of x.
 A. 8
 B. 10
 C. 4
 D. 6
37. If $m:n = 13:11$, find $m^2 - n^2 : (m + n)^2$.
 A. 1:11
 B. 1:13
 C. 1:10
 D. 1:12
38. Compute $110011_2 + 11111_2$.
 A. 1001010₂
 B. 1010010₂
 C. 1000110₂
 D. 1000100₂
39. Simplify $(25)^{-\frac{1}{2}} \times (27)^{-\frac{1}{3}} + (121)^{-\frac{1}{2}} \times (625)^{-\frac{1}{4}}$.
 A. $\frac{34}{55}$
 B. $\frac{9}{11}$
 C. $\frac{14}{5}$
 D. $\frac{3}{275}$
40. The cost of renovating a 6 m square room is ₦540. What is the cost of renovating a 9 m square room?
 A. ₦1 215
 B. ₦ 720
 C. ₦1 620
 D. ₦ 810
41. The sum of the first n positive integers is
 A. $\frac{1}{2}n(n-1)$
 B. $n(n+1)$
 C. $n(n-1)$
 D. $\frac{1}{2}n(n-1)$
42. In a basket, there are 6 grapes, 11 bananas and 13 oranges. If one fruit is chosen at random, what is

the probability that the fruit is either a grape or a banana?

- A. $\frac{6}{30}$
- B. $\frac{5}{30}$
- C. $\frac{17}{30}$
- D. $\frac{11}{30}$

43. Express 123456 to 3 significant figures

- A. 123000
- B. 124000
- C. 125000
- D. 126000

44. Calculate the simple interest on N7500 for 8 years at 5% per annum.

- A. ~~N~~3000
- B. ~~N~~600
- C. ~~N~~300
- D. ~~N~~150

45. The cost of kerosene per litre increases from ~~N~~60 to ~~N~~80. What is the percentage rate of increase?

- A. 42%
- B. 41%
- C. 40%
- D. 25%

46. The probability of picking a letter T from the word OBSTRUCTION is

- A. $\frac{1}{11}$
- B. $\frac{2}{11}$
- C. $\frac{3}{11}$
- D. $\frac{4}{11}$

47. A student spent $\frac{1}{5}$ of his allowances on books, $\frac{1}{3}$ of the remainder on food and kept the rest for contingencies. What fraction was kept?

- A. $\frac{7}{15}$
- B. $\frac{8}{15}$
- C. $\frac{2}{3}$

D. $\frac{4}{5}$

48. Find the capacity in litres of a cylindrical well of radius 1 metre and depth 14 metres.

- A. 44000 litres
- B. 4400 litres
- C. 440 litres
- D. 44 litres

49. If $y = (2x + 1)^3$ find $\frac{dy}{dx}$.

- A. $3(2x + 1)^2$
- B. $6(2x + 1)$
- C. $3(2x + 1)$
- D. $6(2x + 1)^2$

50. In how many ways can a committee of 2 women and 3 men be chosen from 6 men and 5 women?

- A. 100
- B. 30
- C. 200
- D. 50

51. The mean of x, y and 10 is 6. What is the mean of x and y?...

- A. 2
- B. 4
- C. 10
- D. 12

52. Find the stand deviation of 2, 3, 5 and 6.

- A. $\sqrt{\frac{5}{2}}$
- B. $\sqrt{10}$
- C. $\sqrt{\frac{2}{5}}$
- D. $\sqrt{6}$

53. What is the mean deviation of x, 2x, x + 1 and 3x, if their mean 2?

- A. 4
- B. 3
- C. 2
- D. 1

54. Find the radius of a sphere whose surface area is 154cm²

- A. 7.00cm
- B. 3.50cm
- C. 3.00cm

D. 1.75cm

55. The midpoint of P(x, y) and Q(8, 6) is (6, 8). Find x and y
- A. (2, 10)
 - B. (2, 8)
 - C. (2, 12)
 - D. (2, 6)

English

In questions 1 to 20 choose the option nearest in meaning to the word(s) or phrase(s) in *italics*.

1. 'It is very difficult to know who is *entitled* to essential commodities these days.
A. qualified for
B. disqualified from
C. eligible
D. worthy of
2. 'This is a *miniaturized* version of your house' remarked the Architect.
A. compressed
B. abridged
C. decreased
D. smaller.
3. Bola has *sonorous* voice.
A. high-pitched
B. beautiful
C. strong
D. throaty.
4. We have to identify the *protagonists* of the new movement.
A. enemies
B. leading figures
C. opponents
D. Believers
5. The Director left his secret file to the Secretary in *good faith*
A. with proof
B. in anxiety
C. in anticipation
D. in trust.
6. I know you've been *eaves-dropping* all along
A. dropping gently
B. paying attention
C. listening secretly
D. dropping all the leaves.
7. I would like to believe that this brilliant performance of yours is not just a *flash in the pan*.
A. repeated occurrence
B. normal happening
C. flashy happening
D. chance occurrence
8. Our school prefect is too officious and we all hate him because of his behaviour.
A. efficient
B. efficacious
C. over-zealous
D. active
9. The fact that Ruth has inherited her late husband's house is *cold comfort* to her
A. appropriate comfort
B. inadequate comfort
C. no comfort at all
D. wet comfort
10. The advertisement served as a *smoke-screen* for the activities of the company
A. a camouflage
B. a protection
C. an enhancement
D. an exchange.
11. What he received from the visitor was *invaluable*
A. priceless
B. worthless
C. valueless
D. costly.
12. The new leader *hands out an olive branch*.
A. sues for peace
B. gives out branches of the olive tree
C. challenges his opponents to a fight
D. blesses his supporters.
13. When you go to a foreign country to study, you will discover that life there is not a *bed of roses*.
A. as pleasant as one thought
B. a bed with roses
C. an unmitigated disappointment
D. as expected.
14. The woman acted *courageously* when she was attacked by thieves
A. shyly
B. fearlessly

- C. timidly
D. carelessly.
15. We all have both good and bad characteristics. Either is, however, easily *manifested* in times of crisis.
A. demanded
B. highlighted
C. submerged.
D. determined.
16. The tax rebate was a *concession* given to the civil servants.
A. an incentive
B. an allowance
C. work
D. intelligence
17. He was asked to give *copious* examples to appear convincing
A. concrete
B. rigid
C. cogent
D. many
18. Some children *mimic* their teachers
A. imitate
B. mime
C. ridicule
D. tease.
19. Everybody complained of a *lean* harvest last year
A. surplus
B. abundant
C. poor
D. thin.
20. The police *ran the criminal* to earth
A. jailed him
B. knocked him down
C. discovered him
D. buried him
- In questions 21 to 50, choose the option opposite in meaning to the word(s) in italics.**
21. The government is making *concerted* efforts at providing essential social facilities in the rural areas.
A. dissipated
B. unconcerned
C. unsuccessful
D. uncontrolled.
22. The western allies frowned at their enemies' *indiscriminate* bombing of their territory.
A. impartial
B. selective
C. unprovoked
D. divided.
23. We cannot but talk about his *invaluable* contributions to the affairs of the society.
A. worthless
B. costly
C. unrecognized
D. incalculable.
24. The man's health has *deteriorated* in the hospital
A. collapsed
B. improved
C. worsened
D. revived.
25. Ogedegbe was an *intrepid* fighter for human freedom and dignity
A. timid
B. half-hearted
C. fearless
D. undaunted.
26. Ojo's response *infuriated* his wife
A. annoyed
B. pleased
C. surprised
D. confused
27. He accepted a *mundane* task without hesitation.
A. great
B. lowly
C. menial
D. moderate
28. It is a *unique* opportunity for her to demonstrate the reality of her faith
A. strange
B. usual
C. golden
D. unusual
29. The economic situation in the country is obviously *gloomy*
A. encouraging
B. moody
C. unknown
D. regrettable

30. One wonders if the situation will *improve*
 A. brighten
 B. diminish
 C. disintegrate
 D. worsen
31. These two books are *identical*
 A. alike
 B. similar
 C. different
 D. equal
32. The doctor tried to *alleviate* his patients pain
 A. relieve
 B. worsen
 C. kill
 D. cure
33. The instructions on the examination paper are *explicit*
 A. simple
 B. ambiguous
 C. detailed
 D. definite
34. Okoro is an *amateur* wrestler
 A. skilful
 B. good
 C. professional
 D. strong
35. He won the election because he *was able to carry* his people with him.
 A. alienate his people
 B. go out with the people
 C. have the support of the people
 D. fail to provide the people with transport.
36. They often quarreled before leaving the house but *made it up* by the time they returned home
 A. added it up
 B. reconciled
 C. fell out
 D. compounded the problem
37. The effect of rising prices has *snowballed* considerably.
 A. risen
 B. decreased
 C. frozen
 D. grown.
38. We have found that cassava is a *sine qua non* item in our industry.
- A. useful
 B. indispensable
 C. useless
 D. dispensable
39. Science may be a complicated area of learning, but its teaching needs to be *demystified*.
 A. simplified
 B. twisted
 C. made difficult
 D. distorted
40. The stadium was *seething* with people when we entered
 A. empty
 B. noisy
 C. filled
 D. crowded
- In question 66 to 100, choose the word(s) or phrase(s) which best fill(s) the gap(s).**
41. The teacher invited Akpan and ... to his office
 A. myself
 B. me
 C. I
 D. himself.
42. Tsado and his wife are always fighting. The ... to be drawn from this is that they are not happy together.
 A. reference
 B. inference
 C. difference
 D. deference
43. The government has provided new ... for the local government polls.
 A. guard lines
 B. guide lines
 C. guard lines
 D. guidelines.
44. The ... car is the center of attraction.
 A. small racing light-green
 B. racing small light-green
 C. small light-green racing
 D. light-green small racing.
45. Okoro intends to wear his ... dress to the zoo this afternoon.
 A. white cotton new

- B. white new cotton
C. new white cotton
D. cotton white new
46. My uncle is one of the ... of the society.
A. elitists
B. elites
C. elite
D. elitist
47. After Jerry had made the bed, he ... on it.
A. layed
B. laid
C. lied
D. lay
48. Bola broke ... when he heard the news of the arrest of his twin brother.
A. up
B. off
C. away
D. down
49. If it had rained, we would have had to postpone the concert. It didn't rain, so ...
A. we won't have to
B. we would have had to
C. we wouldn't have to
D. we didn't have to.
50. The patient is suffering from ...
A. reumatism
B. rheumatism
C. rhuematism
D. rheumatism.
51. The attitude of my students to ... baffles me
A. pronunciation
B. pronouncation
C. pronunceation
D. pronounceation
52. The story on the lost jewel was quite ...
A. unbelievable
B. unbeliveable
C. unbelieveable
D. unbeleivable
53. When we got to the president's house, we were told that he ... two days earlier.
A. may have left
B. left
C. had left
D. has left.
54. The boy who stole mango was given ... by an eye witness.
A. out
B. in
C. away
D. up.
55. I do not intend to stay for long at the function. I only want to ...
A. put in an appearance
B. put up appearance
C. show up appearance
D. keep up appearance.
56. Do you think all ... not enough to help the investigator?
A. these informations are
B. this information are
C. this information is
D. these information are
57. The trader complained that he... robbed.
A. had been
B. has been
C. was being
D. is being
58. They spent the last holiday with their...
A. sister-in-laws
B. sisters-in-law
C. sisters-in-laws
D. sisters-in-laws
59. The army ran short of ...
A. arm and ammunities
B. arms and ammunities
C. arms and ammunition
D. arm and ammunition.
60. The trader complained that he... robbed.
A. had been
B. has been
C. was being
D. is being
- In each of questions 61 to 90 select the option that best explains the information conveyed in the sentence.**
61. Emeka wished he had started school early
A. Emeka started school early as he wished.
B. Emeka regretted starting school early
C. Emeka regretted not starting school

- D. early
Emeka could have started school early if he had wished.
62. His refusal to help us, dealt a deathblow to our plans.
A. Because he did not help, our plans did not materialize.
B. He helped and our plans materialized
C. He blew our plans to pieces
D. He supported our plans so it worked
63. Whoever would have thought that he would lose the presidential election?
A. Everyone believed that he would lose the election
B. It was commonly believed that he would win the election.
C. He was supposed to lose the election.
D. No one believed that he would win the election.
64. People may not pick flowers in this park.
A. People can pick flowers in this park.
B. People may not wish to pick flowers in this park.
C. People are prohibited from picking flowers in this park.
D. People cannot pick flowers from this park.
65. Tom ought not to have told me.
A. Tom did not tell me but he should.
B. Perhaps Tom was wrong to have told me.
C. Tom told me but it was wrong of him.
D. It was necessary for Tom not to tell me.
66. He can't be swimming all day.
A. It's possible he is not swimming now.
B. It's very likely he is swimming now.
C. He does not have the ability to swim all day.
D. He would not like to swim all day
67. Bolade would make a mess of cooking the rice.
A. It was typical of Bolade to make a mess of things.
B. Bolade cannot cook.
C. Bolade will not cook the rice well.
D. Bolade does not like cooking rice.
68. He is now the megaphone of the people's oppressors.
- A. He now speaks on behalf of the oppressors.
B. He now carries the megaphone when the oppressors speak.
C. He is now the oppressor with the loudest voice.
D. He is the oppressor who now speaks with megaphone.
69. Here is Mr. Pam the tailor.
A. He is the only person known as Mr. Pam, and he is a tailor.
B. He is one of those known as Mr. Pam, and he is a tailor.
C. He is the only tailor, and he is Mr. Pam.
D. He is one of many tailors, and he is Mr. Pam.
70. The mills of God grind slowly, but they grind exceedingly small.
A. God has a mill where every material in life is thoroughly ground.
B. God may seem slow in action, but He adequately rewards every bit of injustice.
C. The world is like a food factory where God takes His time in grinding all raw materials properly.
D. Even though God is never in a hurry, He achieves all things.
71. People are not interested in who rules
A. People who rule are not interested in the ruled
B. The rulers are not indifferent about the people
C. People are indifferent about the rulers
D. People are not ruled by the people they are interested in.
72. My son, who is in the U.S.A., is studying Engineering.
A. My only son is in the U.S.A. studying Engineering.
B. One of my sons is in the U.S.A. studying Engineering.
C. My sons are in the U.S.A. but only one is studying Engineering.
D. Only my son is in the U.S.A. studying Engineering.
73. The stadium had a capacity crowd in spite of the weather.
A. The weather favoured the crowd at the stadium.

- B. The crowd defied the weather to fill the stadium.
C. The stadium was not filled because of the weather.
D. The crowd loved the weather at the stadium.
74. The painting in the museum was beautifully faked.
A. The painting was a good deceptive replica.
B. The painting was well-framed and displayed.
C. The painting was deceptively decorated in the museum.
D. The painting was carefully hung in museum.
75. Ade: Do you mind if I sit down here? Bola: well, yes I do actually.
A. Bola is reluctantly granting the request.
B. Bola is willingly accepting the request.
C. Bola is certainly rejecting the request.
D. Bola is objecting to the request.
76. Fabricated statements published as truths amount to an abuse of press freedom.
A. To make known what is not true is aggressive to press freedom.
B. Publishing formulated untruths is misused press freedom.
C. Unplanned publicized untruths are a misuse of the freedom of the press.
D. Press freedom is based on published untruths.
77. The expected guests arrived at the eleventh hour.
A. The guests arrived at the last minute.
B. The guests arrived a 11 o' clock.
C. The guests arrived at the expected time.
D. The guests arrived eleven hours behind schedule.
78. Benjamin exclaimed, 'If it is Napoleon who said it, it must be right.'
A. Benjamin never believes what Napoleon says.
B. Benjamin does not know what is right unless Napoleon says it.
C. Napoleon always says what Benjamin believes to be right.
D. Benjamin thinks that Napoleon always says the right thing.
79. The crowd in the hall is intimidating.
A. The crowd is frightening.
B. The crowd is angry
C. The crowd is overwhelming.
D. The crowd is riotous.
80. The manager said that the new loaf was the last word in bakery.
A. The loaf was the best ever baked.
B. The loaf was the last to be baked.
C. The loaf was the worst to have been baked.
D. The loaf was the last in the baker's directory.
81. When I visited James I only gave him a congratulatory card on account of his promotion.
A. The only person I visited was James and nobody else.
B. It was only James that I gave a congratulatory card to.
C. All I gave James when I visited him was a congratulatory card.
D. All I did when I visited James was given a congratulatory card.
82. The politician's inflammatory remarks were heavily criticized.
A. The politician's rude remarks were heavily criticized.
B. The politician's provocative remarks were heavily criticized.
C. The politician's bombastic remarks were heavily criticized.
D. The politician's tribalistic remarks were heavily criticized.
83. A. enjoy
B. log
C. toil
D. rejoice
84. The man reasoned that there ought to be a limit to sycophancy.
A. Sycophants need not talk all the time.
B. There can be no favour beyond a reasonable point.
C. Sycophants should know when to grant people's request.

- D. People should know when not to use flattery.
85. Oche's chief idiosyncrasy is a passion for pounded yam.
 A. Oche hates pounded yam.
 B. Oche's chief hates pounded yam.
 C. Oche's chief likes pounded yam.
 D. Oche has a penchant for pounded yam.
86. The politician was pilloried by the press for his inflammatory remarks.
 A. The politician was closely questioned by the press for his remarks.
 B. The politician was criticized by the press for his remarks.
 C. The politician was ridiculed by the press for his remarks.
 D. The politician was snubbed by the press for his remarks.
87. If he went to London, he would see the Queen.
 A. When he goes to London, he will see the Queen.
 B. He did not go to London and did not see the Queen.
 C. He did not see the Queen when he went to London.
 D. He would like to see the Queen when he goes to London.
88. In spite of his humble beginning, Audu now throws his weight around.
 A. Audu is arrogant despite his simple upbringing.
 B. Despite his obvious poverty, Audu is a proud man.
 C. His noble birth notwithstanding, Audu is a corrupt man.
 D. From his poor background, Audu is now a rich man.
89. In many countries, democracy is ostensibly being practiced.
 A. Many countries have effective democracy.
 B. Democracy is indeed being practiced in many countries.
 C. Democracy is apparently practiced in many countries.
 D. Many countries have democracy in practise.
90. Adamu's father is a key figure in that ministry.

- A. Adamu's father's position is essential in the ministry.
 B. The ministry trusts Adamu's father as a central figure.
 C. Adamu's father is a figurehead in the ministry.
 D. Adamu's father keeps the key to the ministry.

Biology

- Which of the following substances is NOT found in urine?
 A. Water
 B. Sodium chloride
 C. Nitrogenous compounds
 D. Calcium chloride E. Nitrogenous salts.
- Starting from the skull end, the vertebrae are arranged in the following order:
 A. axis, atlas, cervical, thoracic and lumbar
 B. atlas, cervical, axis, thoracic and lumbar
 C. atlas, axis, thoracic, cervical and lumbar
 D. atlas, axis, cervical, thoracic and lumbar
 E. atlas, thoracic, cervical axis and lumbar.
- Which of these worms is beneficial to man?
 A. Hookworm
 B. Tapeworm
 C. Roundworm
 D. Earthworm
 E. Guinea worm.
- The mouth part of the housefly are adapted for
 A. lapping and sponging.
 B. sucking and chewing.
 C. piercing and sucking.
 D. chewing and lapping.
 E. biting and chewing.
- Pepsin is a digestive enzyme which breaks
 A. cellulose into glucose molecules
 B. carbohydrates into simple sugars
 C. protein into peptones
 D. fats into glycerol and fatty acids E. sucrose into glucose and fructose.
- Partially digested food ready to leave the stomach is referred to as
 A. chyme
 B. curd
 C. glycogen
 D. paste

- E. roughage
7. Blood platelets are important because they
- are amoeboid and nucleated
 - produce Antitoxins
 - produce antibodies
 - digest harmful bacteria
 - release thrombin for blood clotting.
8. The hereditary material in a cell is known as
- ADP
 - CNS
 - RNA
 - ATP
 - DNA.
9. Incomplete metamorphosis in the
- butterfly
 - grasshopper
 - mosquito
 - housefly
10. Which of the following elements is essential for the formation of haemoglobin?
- Sodium
 - Potassium
 - Calcium
 - Iron.
11. In the mammalian skin, melanin and keratin are contained in the
- sebaceous gland
 - sweat gland
 - subcutaneous layer
 - malpighian layer.
12. Blood circulation in a mammal is said to be double because
- it passes twice through the heart in the complete circuit
 - it moves in both arteries and veins
 - it circulates in both the heart and other organs
 - the heart contains auricles and ventricles
13. The aperture between the left auricle and the left ventricle is guarded by the
- auricular valve
 - tricuspid valve
 - ventricular valve
 - bicuspid valve.
14. The hormone which regulates the amount of glucose in the blood is called
- adrenalin
 - auxin
 - insulin
 - thyroxine.
15. The function of endoplasmic reticulum is
- protein synthesis
 - intracellular transport of materials
 - digestion and destruction of foreign bodies
 - production of energy from glucose.
16. In the life history of a butterfly, destruction of crops is caused by the
- maggot
 - nymph
 - caterpillar
 - pupa.
17. The oxidative part of the respiration process takes place in the
- mitochondria
 - ribosomes
 - endoplasmic reticulum
 - golgilbodies.
18. The vitamin which is important in the formation of the retina pigments is
- vitamin A
 - vitamin B
 - vitamin C
 - vitamin D.
19. The sequence of ear ossicles from the fenestra ovalis is
- malleus, incus and stapes
 - malleus, stapes and incus
 - stapes, incus and mallcus
 - stapes, malleus and incus.
20. The centre for controlling body temperature in the brain is the
- cerebrum
 - cerebellum
 - medulla
 - hypothalamus.
21. Poliomyelitis is an infectious disease caused by
- virus
 - protozoan
 - bacterium
 - fungus.

22. The term 'Thallophyta' refers to
 A. ferns and mosses
 B. algae and fungi
 C. mosses and liverworts
 D. fungi and ferns.
23. If an animal has diastema, it would NOT possess
 A. incisors
 B. canines
 C. premolars
 D. molars
24. If the gall bladder of a mammal is damaged, which of the following will be most seriously affected?
 A. Glycolysis
 B. Digestion of starch
 C. Digestion of fats and oils
 D. Digestion of proteins.
25. In the mammalian respiratory system, exchange of gases occurs in the.
 A. lungs
 B. bronchi
 C. bronchioles
 D. alveoli.
26. What part of the central nervous system is concerned with answering an examination question?
 A. Cerebrum
 B. Cerebellum
 C. Medulla oblongata
 D. Spinal cord.
27. Airspaces are characteristic of seeds or fruits dispersed
 A. birds
 B. water
 C. wind
 D. explosive mechanism.
28. Which of the following relationship involves only one organism?
 A. Saprophytism
 B. Commensalism
 C. Parasitism
 D. Symbiosis.
29. Mineral salts can be absorbed into the roots by
 A. osmosis only
 B. osmosis and diffusion
 C. diffusion and active transport
 D. imbibition only.
30. The pancreatic juice contains the enzymes amylopsin,
 A. pepsin and trypsinogen
 B. rennin and Steapsin
 C. steapsin and trypsinogen
 D. steapsin and ptyalin.
31. Aged erythrocytes are destroyed in the
 A. pancreas
 B. liver
 C. lymph nodes
 D. kidney.
32. The region that controls most of the unconscious processes of a mammalian body is the
 A. cerebellum
 B. cerebrum
 C. spinal cord
 D. medulla oblongata.
33. The urinary tubules of the kidney function through
 A. ultrafiltration and selective reabsorption
 B. active transport and cytoplasmic streaming
 C. osmosis and diffusion
 D. active transport and osmosis
34. A non-renewable alternative source of energy is
 A. hydroelectric power
 B. wind generators
 C. solar panels
 D. nuclear energy
35. The part of the mammalian skin involved in temperature regulation is the
 A. hair follicle
 B. hair papilla
 C. sebaceous gland
 D. sweat gland
36. In rabbits, the chamber of the heart that receives oxygenated blood from the lungs is the
 A. left ventricle
 B. right auricle
 C. left auricle
 D. right ventricle
37. The correct sequence of the movement of urea during urine formation is

- A. convoluted tubule → glomerulus → Henle's loop → Bowman's capsule → collecting tubule.
- B. glomerulus → Bowman's capsule → convoluted tubule → Henle's loop → collecting tubule
- C. convoluted tubule → Bowman's capsule → Henle's loop → glomerulus → collecting tubule
- D. glomerulus → Bowman's capsule → convoluted tubule → Henle's loop → collecting tubule
38. The photosynthetic pigments include
- A. melanin and haemoglobin
- B. chlorophyll and carotenoids
- C. carotenoids and haemoglobin
- D. chloroplasts and cytochromes
39. In mammals, the organ directly on top of the kidney is the
- A. prostate gland
- B. pancreas
- C. thyroid gland
- D. adrenal gland
40. The causative agent of bird flu is a
- A. virus
- B. bacterium
- C. fungus
- D. protozoan
41. The outer layer of the kidney where the Bowman's capsules are found is the
- A. cortex
- B. pelvis
- C. medulla
- D. pyramid
42. The primary structure responsible for pumping blood for circulation through the mammalian circulatory systems is the
- A. veins
- B. right auricle
- C. arteries
- D. left ventricle
43. Circulation of blood to all parts of the body except the lungs is through
- A. the pulmonary artery
- B. systematic circulation
- C. the lymphatic system
- D. pulmonary circulation
44. In the kidney of mammals, the site of ultrafiltration is the
- A. uriniferous tubule
- B. Bowman's capsule
- C. loop of Henle
- D. renal tubule
45. The diseases caused by waterborn pathogens include
- A. gonorrhea and poliomyelitis
- B. typhoid and syphilis
- C. tuberculosis and cholera
- D. typhoid and cholera
46. Which of the following forms about 55% of the volume of the blood in man?
- A. Plasma
- B. Erythrocytes
- C. Leucocytes
- D. Platelets
47. Oestrogen is a hormone that is synthesized in the
- A. anterior pituitary
- B. adrenal cortex
- C. ovaries
- D. testes
48. The stomata of leaves are similar in function to the
- A. trachea of toads
- B. pharynx of humans
- C. scales of fish
- D. spiracle of insects
49. In the male reproductive system of a mammal, sperm is stored in the
- A. vas deferens
- B. urethra
- C. epididymis
- D. seminiferous tubules
50. The two halves of the pelvic girdle are joined together at the
- A. pubic symphysis
- B. ilium
- C. pubis
- D. obturator foramen
51. The first terrestrial vertebrates evolved from
- A. Pisces
- B. Reptilia
- C. Amphibia
- D. mammalian

52. During ovulation, an egg is released from the
 A. corpus luteum
 B. ovarian funnel
 C. Graafian follicle
 D. fallopian tube
53. In the mammalian kidney, the Bowman's capsule is located in the
 A. ureter
 B. pelvis
 C. cortex
 D. medullar
54. The order of passage of food in the digestive system is
 A. ileum → caecum → large intestine → rectum
 B. ileum → colori → caecum → rectum
 C. large intestine → ileum → caecum → rectum
 D. colon → caecum → ileum → rectum.
55. Farmers practices crop rotation because it
 A. helps to prevent soil erosion
 B. allows two crops to be planted at the same time
 C. helps to conserve soil fertility
 D. is an alternative to shifting cultivation
56. Which of the following diseases can be caused by a bacterium?
 A. ringworm
 B. poliomyelitis
 C. malaria
 D. syphilis.
57. Viruses are considered to be living organisms because they
 A. possess transmittable characters
 B. move from one place to another
 C. respond to stimulation
 D. ingest food materials
58. Which of the following food substances will produce a brick-red colour when warmed with Benedict's solution
 A. Glucose
 B. Starch
 C. Egg white
 D. Maltose.
59. The villus in the small intestine is significant because it
 A. increases the surface area for absorption
 B. increases the surface area for digestion
 C. assists in mixing digested food
 D. assists in filtering undigested food.
60. The part of the ear that is concerned with balance is the
 A. semi-circular canals
 B. incus
 C. malleus
 D. Bastachian tube.
61. The part of the ear that is concerned with balance is the
 A. semi-circular canals
 B. incus
 C. malleus
 D. Bastachian tube.
61. In the earthworm, the cocoon is secreted by the
 A. chaeta
 B. prostomium
 C. peristomium
 D. clitellum.
62. Which of the following is a plant excretory product?
 A. Oil
 B. Cytokinin
 C. Resin
 D. Amino acids.
63. The crown of the mammalian tooth is covered with
 A. cement
 B. dentine
 C. caries
 D. enamel.
64. Which of the following characters is NOT sex-linked?
 A. River blindness
 B. Baldness.
 C. Haemophilia
 D. Colour Blindness
65. The major site of photosynthesis in the leaf is the
 A. palisade parenchyma
 B. mesophyll parenchyma
 C. upper epidermis
 D. lower epidermis.

66. The most effective method of dealing with non-biodegradable pollution is by
 A. burying
 B. dumping
 C. incineration
 D. recycling
67. The membrane surrounding the vacuole in a plant cell is called the
 A. plasmalemma
 B. tonoplast
 C. nuclear membrane
 D. endoplasmic reticulum.
68. Sex-linked genes are located on
 A. X-and Y-chromosomes
 B. homologous chromosomes
 C. X- chromosomes
 D. Y- chromosome.
69. The anaerobic fermentation of a glucose molecule yields
 A. pyruvic acid and alcohol
 B. 38 ATP molecules
 C. water and carbon (IV) oxide
 D. 2ATP molecules and alcohol
70. The two key cations involved in the action potential of nervous transmissions are
 A. Mg^{2+} and K^{+}
 B. Na^{+} and Fe^{2+}
 C. Fe^{2+} and Mg^{2+}
 D. Na^{+} and K^{+}
71. The gall bladder of a mammal has a duct connected to the
 A. duodenum
 B. liver
 C. pancreas
 D. small intestine
72. If a DNA strand has a base sequence TCA, its complementary strand must be
 A. ATG
 B. GAT
 C. AGT
 D. TAG
73. In mammalian males, the excretory and reproductive system share the
 A. ureter
 B. testes
 C. vas deferens
 D. urethra
74. The response of plants to external stimuli in a non-directional manner is known as
 A. nastic movement
 B. phototropism
 C. tactic
 D. geotropism
75. A sex-linked character cannot be passed on directly from
 A. father to son
 B. mother to daughter
 C. mother to son
 D. father to daughter.
76. The biological association that contributes directly to succession in a community is
 A. competition
 B. predation
 C. parasitism
 D. commensalism.
77. The part of the brain that controls body posture in mammals is the
 A. thalamus
 B. cerebrum
 C. spinal cord
 D. cerebellum
- Physics**
1. A man of mass 50kg ascends a flight of stairs 5m high in 5 seconds. If acceleration due to gravity is 10 m s^{-2} , the power expended is
 A. 100W
 B. 300W
 C. 250W
 D. 400W
 E. 500W
2. The refractive index of a liquid is 1.5. If the velocity of light in vacuum is $3.0 \times 10^8\text{ m s}^{-1}$, the velocity of light in the liquid is
 A. $1.5 \times 10^3\text{ m s}^{-1}$
 B. $2.0 \times 10^3\text{ m s}^{-1}$
 C. $3.0 \times 10^3\text{ m s}^{-1}$
 D. $4.5 \times 10^3\text{ m s}^{-1}$
 E. $9.0 \times 10^3\text{ m s}^{-1}$
3. If the relative density of a metal is 19, what will be the mass of 20 cm^3 of the metal when immersed in water?
 A. 380g
 B. 400g
 C. 360g

- D. 39g
E. 180g
4. Four equal resistors R1, R2, R3 and R4 are connected in series as shown in Fig 9 below. V1, V2 and V3 are voltmeters connected as indicated. Which of the following relations is CORRECT?
A. $V1 = V3 = V2$.
B. $V1 = 2V2 = V3$
C. $V1 = \frac{1}{2} V3 = V2$
D. $V1 - V3 = V2$
E. $V2 - V1 = V3$
5. A simple pendulum with a period of 2.0s has its length doubled. Its new period is
A. 1.00 s
B. 1.41 s
C. 0.35 s
D. 2.83 s
E. 4.00 s
6. Which of the following represents the correct precision if the length of a piece of wire is measured with a metre rule?
A. 35 mm
B. 35.0 mm
C. 35.00 mm
D. 35.01 mm
7. If the refractive index of glass is 1.5, what is the critical angle at the air-glass interface?
A. $\sin^{-1} \frac{1}{2}$
B. $\sin^{-1} \frac{2}{3}$
C. $\sin^{-1} \frac{3}{4}$
D. $\sin^{-1} \frac{8}{9}$
8. In which of the following are the substances arranged in descending order of their thermal conductivities?
A. Copper, steel, glass
B. Steel, copper, glass
C. Steel, glass, copper
D. Copper, glass, steel.
9. A radioactive sample initially contains N atoms. After three half-lives the number of atoms that have disintegrated is
A. N 8
B. 3N 8
C. 5N 8
D. 7N 8
10. A cell of internal resistance 2 ohms supplies current to a 6-ohm resistor. The efficiency of the cell is
A. 12.0 %
B. 25.0 %
C. 33.3 %
D. 75.0 %
11. Which of the following is a fundamental unit?
A. Newton
B. Watt
C. Joule
D. Second
12. Which of the following obeys Ohm's Law?
A. Glass
B. Diode
C. All electrolytes
D. All metals.
13. When an atom loses or gains a charge, it becomes
A. An electron
B. An ion
C. A neutron
D. A proton.
14. If the refractive index of a medium in air is 2.0, what is the critical angle for this medium?
A. 300
B. 420
C. 450
D. 500
15. Which of the following factors has no effect on e.m.f. of a primary cell?
A. Temperature
B. Size of the cell
C. Nature of the plates
D. Nature of the electrolyte.
16. The external and internal diameters of a tube are measured as (32 ± 2) mm and (21 ± 1) mm respectively. Determine the percentage error in the thickness of the tube.
A. 27
B. 14%
C. 9%
D. 3%
17. Which of the following quantities has the same unit as the watt?
A. Force x time
B. Force x distance
C. Force x acceleration

- D. Force x velocity
18. 200g of water at 90°C is mixed with 100g of water at 30°C. What is the final temperature?
 A. 50°C
 B. 60°C
 C. 70°C
 D. 80°C
19. When the volume of a given mass of gas is halved and its temperature doubled, the pressure
 A. Remains constant
 B. Increases by a factor of 4
 C. Increases by a factor of 3
 D. Decreases by a factor of 4
20. A motor vehicle is brought to rest from a speed of 15 ms⁻¹ in 20 seconds. Calculate the retardation.
 A. 0.75 ms⁻²
 B. 1.33 ms⁻²
 C. 5.00 ms⁻²
 D. 7.50 ms⁻²
21. To produce a parallel beam of light from a concave mirror, the distance at which the lamp should be placed from the mirror is equal to
 A. the focal length
 B. two times the focal length
 C. the distance of the image
 D. two times the radius of curvature.
22. An object is placed directly below a glass block of thickness 3.0 cm. Calculate the lateral displacement of the object if the refractive index of the glass is 1.5
 A. 1.0 cm
 B. 1.5 cm
 C. 2.0 cm
 D. 2.5 cm
23. Temperature is a measure of the
 A. Quantity of heat transferred into the molecules of an object
 B. Mean kinetic energy of any molecules of the object
 C. Kinetic energy of any individual molecules of the substance.
 D. Amount of work done by the molecules of the object.
24. The energy contained in a wire when it is extended by 0.02m by a force of 500 N is
 A. 5J
- B. 10 J
 C. 10 J
 D. 104 J
25. A parallel capacitor has a common plate area of 5 x 10⁻⁸m² and plate separation of 2 x 10⁻³m. Assuming free space, what is the capacitance?
 A. 2.25 x 10⁻¹⁷F
 B. 4.50 x 10⁻¹⁷F
 C. 2.25 x 10⁻¹⁶F
 D. 4.50 x 10⁻¹⁶F
26. An electric kettle with negligible heat capacity is rated at 2000 W. If 2.0kg of water is put in it, how long will it take the temperature of water to rise from 20°C to 100°C?
 A. 420s
 B. 336s
 C. 168s
 D. 84s.
27. The displacement d produced in a glass block of thickness t and refractive index n when an object is viewed through it is
 A. $t-n$
 B. $t(1 + 1/n)$
 C. $t(1 - 1/n)$
 D. $t(1/n - 1)$
28. A car of mass 800kg attains a speed of 25ms⁻¹ in 20 seconds. The power developed in the engine is
 A. 1.25 x 10⁴ W
 B. 2.50 x 10⁴W
 C. 1.25 x 10⁶ W
 D. 2.50 x 10⁶ W
29. The inner diameter of a small test tube can be measured accurately using a
 A. Micrometer screw gauge
 B. Pair of dividers
 C. Metre rule
 D. Pair of vernier calipers.
30. A 3000 W electric cooker is to be used on a 200V mains circuit. Which of the fuses below can be used safely with the cooker?
 A. 2A
 B. 5A
 C. 10A
 D. 20A
31. A catapult used to hold a stone of mass 500g is extended by 20cm with an applied force f . If the

- stone leaves with a velocity of 40 ms^{-1} , the value of F is
- $4.0 \times 10^4 \text{ N}$
 - $4.0 \times 10^3 \text{ N}$
 - $2.0 \times 10^3 \text{ N}$
 - $4.0 \times 10^2 \text{ N}$
32. If the force and the velocity on a system are each reduced simultaneously by half, the power of the system is
- Doubled
 - Constant
 - Reduced to a quarter
 - Reduced by half.
33. The velocity ratio of a machine is 5 its efficiency is 75%. What effort would be needed to lift a load of 150 N with the machine?
- 50 N
 - 40 N
 - 30 N
 - 20 N
34. A boy observes a piece of stone at the bottom of a river 6.0 m deep. If he looks from the surface of the river, what is the apparent distance of the stone from him.
- 4.5 m
 - 5.0 m
 - 5.5 m
 - 8.0 m
35. A cell of internal resistance 0.01 ohms can be measured accurately using the
- Ohm-meter
 - Potentiometer
 - Electroscope
 - Metre bridge
36. A radio is operated by eight cells each of e.m.f. 2.0 V connected in series. if two of the cells are wrongly connected, the net e.m.f. of the radio is
- 16 V
 - 12 V
 - 10 V
 - 8 V
37. A certain radioactive source emits radiations that were found to be deflected by both magnetic and electric fields. The radiations are
- x-rays
 - beta rays
 - gamma rays
 - ultra-violet rays
38. The thermometric substance of an absolute thermometer is
- Alcohol
 - Mercury
 - Helium
 - Platinum
39. A cell of internal resistance r supplies current to a 6.0 Ohms resistor and its efficiency is 75%. Find the value of r
- 4.5 Ohms
 - 1.0 Ohms
 - 8.0 Ohms
 - 2.0 Ohms
40. When connected to a mains 250 V , the fuse rating in the plug of an electric device of 1 kW is
- 5 A
 - 4 A
 - 3 A
 - 2 A
41. Water is poor thermometric liquid because it
- Wets glass
 - Has low vapour pressure
 - Is opaque
 - Is a poor conductor
42. magnification produced by a convex lens of focal length f ?
- $u + f$
 - $u - f$
 - $v - 1$
 - $v + 1$
43. The pitch of a sound note depends on
- Timbre
 - Harmonics
 - Quality
 - frequency
44. I. Jet-propelled aircraft
II. Rocket propulsion
III. The recoil of a gun
IV. A person walking
Which of the above is based on Newton's third law of motion?
- I, II, III and IV
 - I and III only
 - I and II only
 - I, II and III only

45. Given three capacitors $0.3\mu\text{F}$, $0.5\mu\text{F}$ and $0.2\mu\text{F}$, the joint capacitance when arranged to give minimum capacitance is
- A. $0.3\mu\text{F}$
 - B. $1.0\mu\text{F}$
 - C. $0.1\mu\text{F}$
 - D. $0.5\mu\text{F}$